In-class Exercise #1
Basic Analytics of Hospital Waiting Time

Scenario

A hospital has received complaints that patients are waiting too long before seeing a doctor. Your team has been asked to analyze the historical consultation data and provide recommendations.

Business Problem

Long patient waiting times reduce customer satisfaction and hospital efficiency.

Business Goal

Use the available data to identify factors associated with waiting time and provide evidence-based recommendations.

Dataset

One Excel file containing daily patient consultation records.

- Financial (insurance) type
- Date of visit
- Arrival time
- Other related attributes

Business Questions

Q1. Does the financial (insurance) type affect patient waiting time? Explain your findings using appropriate charts.

Q2. Does the day of the week affect patient waiting time? Explain your findings using appropriate charts.

Your Tasks

1. Perform Exploratory Data Analysis (EDA).
2. Understand the dataset (data types, missing values, unusual values).
3. Create appropriate visualizations.
4. Interpret every chart from a business perspective.

Possible Tools

- Excel
- Python (Pandas, Matplotlib, Seaborn)
- Power BI
- Tableau
- Other analytics tools

Possible Charts

- Bar Chart
- Box Plot
- Histogram
- Heatmap
- Scatter Plot
- Other Types of Charts

Submit (20 points)

1) StudentID_Inclass#1.pdf => Include charts, analysis and business interpretation
2) Source files (.xlsx, .ipynb, .pbix, etc.) => Make sure P'Jame(TA) can open and run your files.

Think Like a Data Analyst

Creating charts is not the goal. The goal is to answer business questions and support better decision-making.

End-of-Class Discussion

Can we answer the following questions using only this Excel file?

- Which doctor has the longest consultation time?
- Are patients satisfied?
- Did weather affect arrivals?
- What complaints appeared on social media?

These questions require multiple data sources. This motivates Data Warehousing, ETL, and Complex Data Analytics, which will be introduced in the next class.